Supporting Information

Molar Mass Determination for Small and Large Molecules using Diffusion-Ordered Spectroscopy

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1) Tables:

Table S1. Molar masses (in g/mol) of PS-, PMMA, PI, P2VP and PEO homopolymers used for the DOSY calibration.

Polymer	Sample	M_p	M_w	M_n	PDI
PS	ps1	1306	1220	1120	1.09
	ps2	3500	3510	3350	1.05
	ps3	8680	8670	8420	1.03
	ps4	19700	19100	18200	1.05
	ps5	34800	34000	32700	1.04
	ps6	66000	62500	59300	1.05
	ps7	130000	125000	120000	1.04
	ps8	304000	294000	282000	1.04
	ps9	552000	554000	532000	1.04
	ps10	1240000	1170000	1100000	1.06
PMMA	pmma1	2160	2130	1920	1.11
	pmma2	5980	5930	5590	1.06
	pmma3	12900	12800	12300	1.04
	pmma4	22800	21200	18700	1.13
	pmma5	41400	40300	38100	1.06
	pmma6	88500	86700	83700	1.04
	pmma7	202000	187000	175000	1.07
	pmma8	520000	517000	501000	1.03
	pmma9	988000	936000	850000	1.10
PI	pi1	1030	1070	944	1.13
	pi2	2200	2190	2030	1.08
	pi3	4620	4610	4450	1.04
	pi4	12100	12000	11800	1.02
	pi5	23500	23600	23300	1.01
	pi6	42300	42500	42000	1.01
	pi7	111000	111000	110000	1.01
	pi8	226000	231000	227000	1.02
	pi9	443000	443000	429000	1.03
	pi10	1180000	1090000	1010000	1.08
P2VP	p2vp1	1020	1050	867	1.21
	p2vp2	2450	2490	2180	1.14
	p2vp3	5780	5790	4630	1.25
	p2vp4	11500	12200	11500	1.06
	p2vp5	19600	20500	19600	1.04
	p2vp6	42800	43100	40500	1.06
	p2vp7	118000	120000	113000	1.06
	p2vp8	243000	249000	232000	1.07
	p2vp9	539000	535000	488000	1.10
	p2vp10	1160000	1200000	1040000	1.15
PEO	peo1	1030	1020	978	1.04

peo2	3450	3450	3340	1.03
peo3	6530	6200	5860	1.06
peo4	12600	11400	10300	1.11
peo5	26100	25800	22100	1.17
peo6	42700	40100	30700	1.31
peo7	99000	103000	91800	1.12
peo8	217000	220000	196000	1.12
peo9	504000	504000	461000	1.09
peo10	1180000	1200000	1110000	1.08

Table S2. Molar masses (in g/mol) of small molecules used for the DOSY calibration. The samples are separated according to their shapes: CS=compact spheres, DSE=dissipated spheres and ellipsoids, ED=expanded discs (nomenclature according to ref (1,2)).

Shape	Sample	M [g/mol]
CS	Cyclopentane	70
	THF	72
	TMS	88
	Methyl-*tert*-butylether (MTB)	88
	Tetramethylbutane (TMB)	114
	Adamantane	136
	$N(SiMe_3)_3$	234
	$Si(SiMe_3)_4$	321
DSE	Toluene	92
	Diisopropylether	102
	Indene	116
	Naphthalene	128
	1,3-Indandione	146
	2-Phenylpyridine	155
	Tetramethoxypropane	164
	Diphenylacetylene	178
	Diphenylsulfoxide	202
	1-Phenylnaphthalene	204
	Triphenylmethane	244
	Tri-(o-tolyl)-phosphine	304
	Hexaphenyltrisiloxane	595
	BINAP	623
ED	Anthracene	178
	Acridine	179
	9-Methylanthracene	192
	Pyrene	202
	Anthrachinone	208
	Triphenylene	228
	3,4-Dihydrobenzophenanthrene	281
	Tetraphenylnaphthalene	433

Table S3. Pulsed field gradient conditions for all polymers and solvents: Gradient pulse length given as little delta (LD) and diffusion delay as big delta (BD) for the convection compensated pulse sequence dstebpgp3s

Sample	Gradients	$CDCl_3$	C_6D_6	CD_2Cl_2	THF	Toluene	Acetone
ps1	LD/ms	2.0	2.0	1.6	2.0	2.0	1.6
	BD/ms	76.73	76.73	77.27	76.73	76.73	57.27
ps2	LD/ms	2.0	2.0	1.6	2.0	2.0	1.6
	BD/ms	76.73	76.73	77.27	76.73	76.73	57.27
ps3	LD/ms	2.6	2.6	2.4	2.6	2.6	2.0
	BD/ms	95.93	95.93	76.2	95.93	95.93	76.73
ps4	LD/ms	3.2	3.2	3.0	3.2	3.2	2.6
	BD/ms	95.13	95.13	95.4	95.13	95.13	75.93
ps5	LD/ms	3.8	3.8	3.6	3.8	3.8	3.0
	BD/ms	94.33	94.33	94.6	94.33	94.33	75.4
ps6	LD/ms	4.2	4.2	4.0	4.2	4.2	-
	BD/ms	113.8	113.8	94.07	113.8	113.8	-
ps7	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	111.93	91.93	141.93	91.93	-	-
ps8	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	191.93	191.93	251.93	191.93	-	-
ps9	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	241.93	251.93	241.93	-	-
ps10	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	391.93	341.93	391.93	-	-
Sample	**Gradients**	**$CDCl_3$**	**C_6D_6**	**CD_2Cl_2**	**THF**	**DMSO**	-
pmma1	LD/ms	2.0	2.0	1.6	2.0	3.6	-
	BD/ms	76.73	76.73	77.27	76.73	74.6	-
pmma2	LD/ms	2.0	2.0	2.0	2.2	3.8	-
	BD/ms	76.73	76.73	76.73	76.47	94.33	-
pmma3	LD/ms	2.6	2.6	2.4	2.6	4.0	-
	BD/ms	95.93	95.93	96.2	95.93	94.07	-
pmma4	LD/ms	3.2	3.2	3.0	3.2	4.6	-
	BD/ms	95.13	95.13	95.4	95.13	113.27	-
pmma5	LD/ms	3.8	3.8	3.6	3.8	5.2	-
	BD/ms	94.33	94.33	94.6	94.33	122.47	-
pmma6	LD/ms	4.2	4.2	4.0	4.2	5.6	-
	BD/ms	113.8	113.8	94.07	113.8	141.93	-
pmma7	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	141.93	141.93	141.93	91.93	-	-
pmma8	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	291.93	251.93	191.93	-	-
pmma9	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	391.93	391.93	341.93	341.93	-	-
Sample	**Gradients**	**$CDCl_3$**	**C_6D_6**	**CD_2Cl_2**	**THF**	-	-
pi1	LD/ms	1.6	2.0	1.6	1.8	-	-
	BD/ms	77.27	76.73	77.27	77.0	-	-
pi2	LD/ms	2.0	2.0	1.8	2.0	-	-

	BD/ms	77.0	76.73	77.0	76.73	-	-
pi3	LD/ms	2.4	2.4	2.4	2.6	-	-
	BD/ms	96.2	96.2	96.2	95.93	-	-
pi4	LD/ms	3.0	3.2	2.6	3.2	-	-
	BD/ms	95.4	95.13	95.93	95.13	-	-
pi5	LD/ms	3.6	4.0	3.6	3.8	-	-
	BD/ms	94.6	94.07	94.6	94.33	-	-
pi6	LD/ms	4.2	4.4	4.2	4.2	-	-
	BD/ms	113.8	113.53	113.8	113.8	-	-
pi7	LD/ms	4.6	5.2	4.6	4.8	-	-
	BD/ms	143.27	142.47	113.27	143.0	-	-
pi8	LD/ms	5.6	5.6	-	5.6	-	-
	BD/ms	191.93	191.93	-	161.93	-	-
pi9	LD/ms	5.6	5.6	-	5.6	-	-
	BD/ms	291.93	291.93	-	261.93	-	-
pi10	LD/ms	5.6	5.6	-	5.6	-	-
	BD/ms	491.93	491.93	-	441.93	-	-
Sample	**Gradients**	**C_6D_6**	**CD_2Cl_2**	**THF**	**MeOD**	-	-
p2vp1	LD/ms	1.8	1.6	1.6	1.8	-	-
	BD/ms	77.0	77.27	77.27	77.0	-	-
p2vp2	LD/ms	2.0	1.6	1.8	2.0	-	-
	BD/ms	76.73	77.27	77.0	76.73	-	-
p2vp3	LD/ms	2.4	2.0	2.0	2.4	-	-
	BD/ms	96.2	96.73	96.73	96.2	-	-
p2vp4	LD/ms	2.8	2.4	2.4	2.8	-	-
	BD/ms	95.67	96.2	96.2	95.67	-	-
p2vp5	LD/ms	3.4	2.8	2.8	3.2	-	-
	BD/ms	94.87	105.67	105.67	105.13	-	-
p2vp6	LD/ms	4.0	3.4	3.4	3.8	-	-
	BD/ms	114.07	114.87	114.87	114.33333	-	-
p2vp7	LD/ms	4.6	4.0	4.0	4.4	-	-
	BD/ms	123.27	124.07	124.07	123.53	-	-
p2vp8	LD/ms	5.6	5.6	5.6	-	-	-
	BD/ms	141.93	141.93	141.93	-	-	-
p2vp9	LD/ms	5.6	5.6	5.6	-	-	-
	BD/ms	191.93	191.93	191.93	-	-	-
p2vp10	LD/ms	5.6	5.6	5.6	-	-	-
	BD/ms	341.93	341.93	341.93	-	-	-
Sample	**Gradients**	**MeOD**	**D_2O**	**CD_3CN**	**THF**	**Acetone**	-
peo1	LD/ms	2.0	2.6	1.6	2.0	1.6	-
	BD/ms	76.73	75.93	77.27	76.73	57.27	-
peo2	LD/ms	2.0	2.6	1.6	2.0	1.6	-
	BD/ms	76.73	75.93	77.27	76.73	57.27	-
peo3	LD/ms	2.6	3.2	2.4	2.4	2.0	-
	BD/ms	95.93	95.13	76.2	96.2	76.73	-
peo4	LD/ms	3.2	3.8	3.0	3.0	2.6	-

	BD/ms	95.13	94.33	95.4	95.4	75.93	-
peo5	LD/ms	3.8	4.6	3.6	3.6	3.0	-
	BD/ms	94.33	93.27	94.6	94.6	75.4	-
peo6	LD/ms	4.2	5.2	4.0	4.0	-	-
	BD/ms	113.8	112.47	94.07	114.07	-	-
peo7	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	141.93	241.93	91.93	141.93	-	-
peo8	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	341.93	161.93	191.93	-	-
peo9	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	491.93	241.93	291.93	-	-
peo10	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	491.93	641.93	441.93	491.93	-	-

Table S4a. Parameters b (Flory parameter) and c of the power scaling $D=c\cdot M^{-b}$ determined via DOSY and the dynamic viscosities (in Pa·s) of the solvents for small molecules of different shapes at T=298 K: compact spheres (CS), dissipated spheres and ellipsoids (DSE) and expanded discs (ED) (parameters b and c are calculated from diffusion data of ref (1) reproduced with permission of the Royal Society of Chemistry and ref (2) reproduced with permission of John Wiley and Sons, respectively). The dynamic viscosities are related to the references and temperature dependences of Table S5.

Sample	THF-d8	C_6D_6	$CDCl_3$	CD_2Cl_2	Toluene-d8	DMSO-d6	C_6D_{12}
CS / b	0.50087	0.46669	0.4635	0.4837	0.5065	0.61984	0.58898
CS / c	1.8988E-8	1.562E-8	1.5319E-8	2.1503E-8	1.7992E-8	1.0117E-8	1.9022E-8
CS / η	0.0005011	0.0006381	0.0005444	0.0004137	0.000619	0.002185	0.0009597
DSE / b	0.61287	0.62975	0.57938	0.56926	0.61902	0.72055	0.7025
DSE / c	3.4413E-8	3.5605E-8	2.6851E-8	3.3222E-8	3.1356E-8	1.7684E-8	3.8083E-8
DSE / η	0.0005011	0.0006381	0.0005444	0.0004137	0.000619	0.002185	0.0009597
ED / b	0.78013	0.76497	0.61563	0.63314	0.8235	0.89594	1.0642
ED / c	8.8757E-8	7.6004E-8	3.5927E-8	5.1783E-8	9.8044E-8	4.6005E-8	2.8854E-7
ED / η	0.0005011	0.0006381	0.0005444	0.0004137	0.000619	0.002185	0.0009597

Table S4b. Parameters b (Flory parameter) and c of the power scaling $D=c\cdot M^{-b}$ determined via DOSY and the dynamic viscosities (in Pa·s) of the solvents for PS, PMMA, PI, P2VP and PEO at T=298.15 K. The dynamic viscosities are related to the references and temperature dependences of Table S5.

Sample	THF-d8	C_6D_6	$CDCl_3$	CD_2Cl_2	Toluene-d8	Acetone-d6
PS / b	0.5604	0.56041	0.56239	0.56162	0.53106	0.45209
PS / c	3.1796E-8	2.5445E-8	2.9966E-8	3.92E-8	2.1376E-8	2.0934E-8
PS / η	0.000501	0.000636	0.000544	0.000413	0.0006179	0.000326
	THF-d8	**C_6D_6**	**$CDCl_3$**	**CD_2Cl_2**	**DMSO-d6**	-
PMMA/b	0.5624	0.56913	0.58372	0.58736	0.44424	-
PMMA/c	3.6793E-8	2.9624E-8	3.8316E-8	5.1798E-8	3.1205E-9	-
PMMA/η	0.000501	0.000636	0.000544	0.000413	0.00218	-
	THF-d8	**C_6D_6**	**$CDCl_3$**	**CD_2Cl_2**	-	-
PI / b	0.60811	0.61614	0.62721	0.57317	-	-
PI / c	4.672E-8	4.0871E-8	5.0317E-8	4.1675E-8	-	-
PI / η	0.000501	0.000636	0.000544	0.000413	-	-
	THF-d8	**C_6D_6**	**CD_2Cl_2**	**MeOD**	-	-
P2VP/b	0.5049	0.48883	0.53396	0.4789	-	-
P2VP/c	1.8199E-8	1.2484E-8	2.6449E-8	1.1295E-8	-	-

P2VP/η	0.000501	0.000636	0.000413	0.000602	-	-
	THF-d8	**CD_3CN**	**D_2O**	**MeOD**	**Acetone-d6**	-
PEO / b	0.58062	0.59613	0.5723	0.57021	0.47448	-
PEO / c	3.3196E-8	5.225E-8	1.3268E-8	2.5609E-8	2.1208E-8	-
PEO / η	0.000501	0.000357	0.001098	0.000602	0.000326	-

Table S4c. Parameters b (Flory parameter) and c of the power scaling $D=c \cdot M^{-b}$ determined via DOSY and the dynamic viscosities (in Pa·s) of the solvents for PMMA in C_6D_6 and PEO in D_2O at variable temperatures in Kelvin. The dynamic viscosities are related to the references and temperature dependences of Table S5.

Sample	**298.15 K**	**305.15 K**	**312.15 K**	**319.15 K**	**326.15 K**	**333.15 K**
PMMA/b	0.52734	0.52996	0.52552	0.53221	0.51415	0.51461
PMMA/c	2.0111E-8	2.3805E-8	2.58E-8	3.0718E-8	2.9217E-8	3.2651E-8
PMMA/η	0.000636	0.0005771	0.0005253	0.0004801	0.0004405	0.0004057
Sample	**298.15 K**	**303.15 K**	**313.15 K**	**323.15 K**	**333.15 K**	-
PEO / b	0.5723	0.55987	0.56339	0.55747	0.55169	-
PEO / c	1.3268E-8	1.4378E-8	1.9152E-8	2.3712E-8	2.7776E-8	-
PEO / η	0.001098	0.0009738	0.0008001	0.0006653	0.0005595	-

Table S5. Temperature dependences of the dynamic viscosities in Pa·s calculated with $\mathbf{log}(\boldsymbol{\eta}) = \frac{\boldsymbol{d}}{\boldsymbol{T}} + \boldsymbol{e}$.

Solvent	**d**	**e**	**literature**
THF-d8	404.06770	-4.65605	ref. (3)
C_6D_6	555.68004	-5.05978	ref. (3)
$CDCl_3$	381.31056	-4.54363	ref. (4)
CD_2Cl_2	428.82173	-4.82233	ref. (5)
Toluene-d8	442.06333	-4.69177	ref. (6,7)
DMSO-d6	756.47082	-5.19904	ref. (3)
C_6D_{12}	653.21399	-5.20988	ref. (8)
MeOD	549.48078	-5.06149	ref. (5)
D_2O	810.387038	-5.68473	ref. (9,10)
CD_3CN	388.56288	-4.74954	ref. (3)
Acetone-d6	376.24429	-4.749	ref. (3)

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2) Figures:

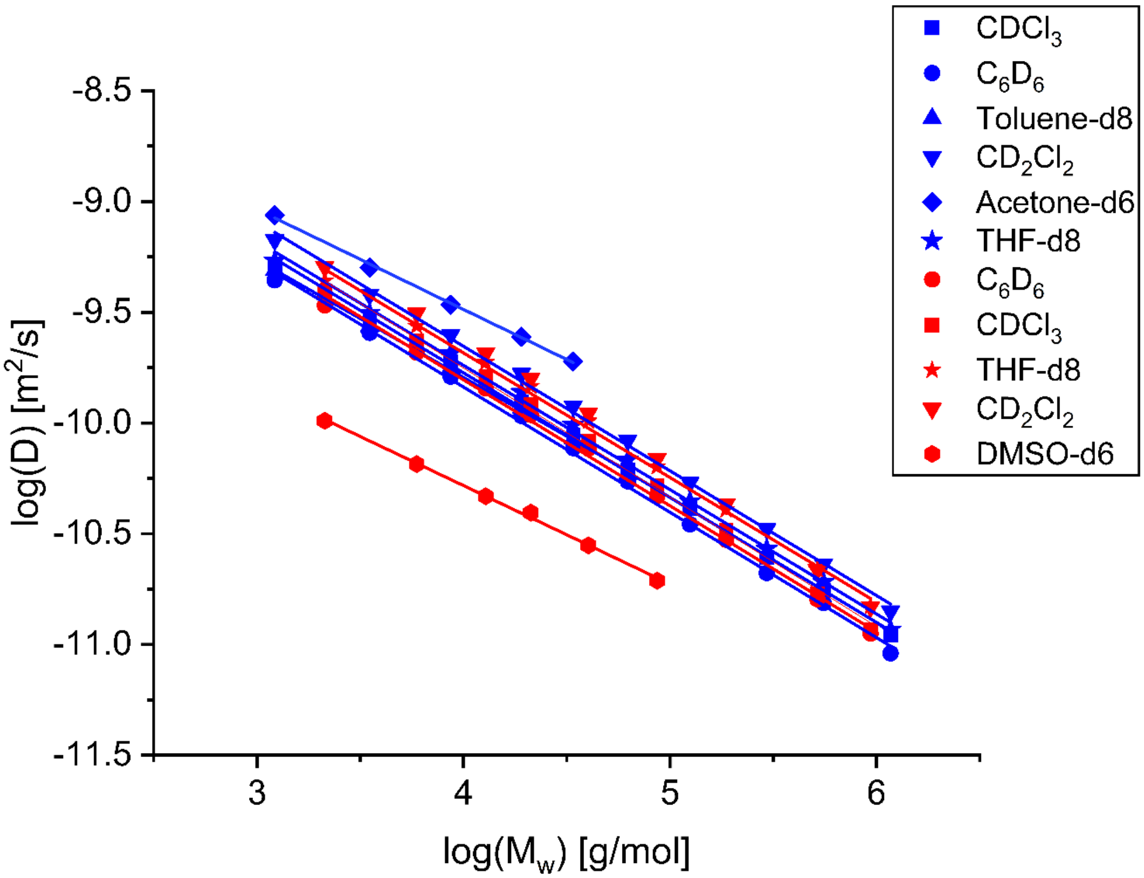

Figure S1. Diffusion coefficients of PS (blue symbols) and PMMA (red symbols) in dependence of the molar mass for different solvents.

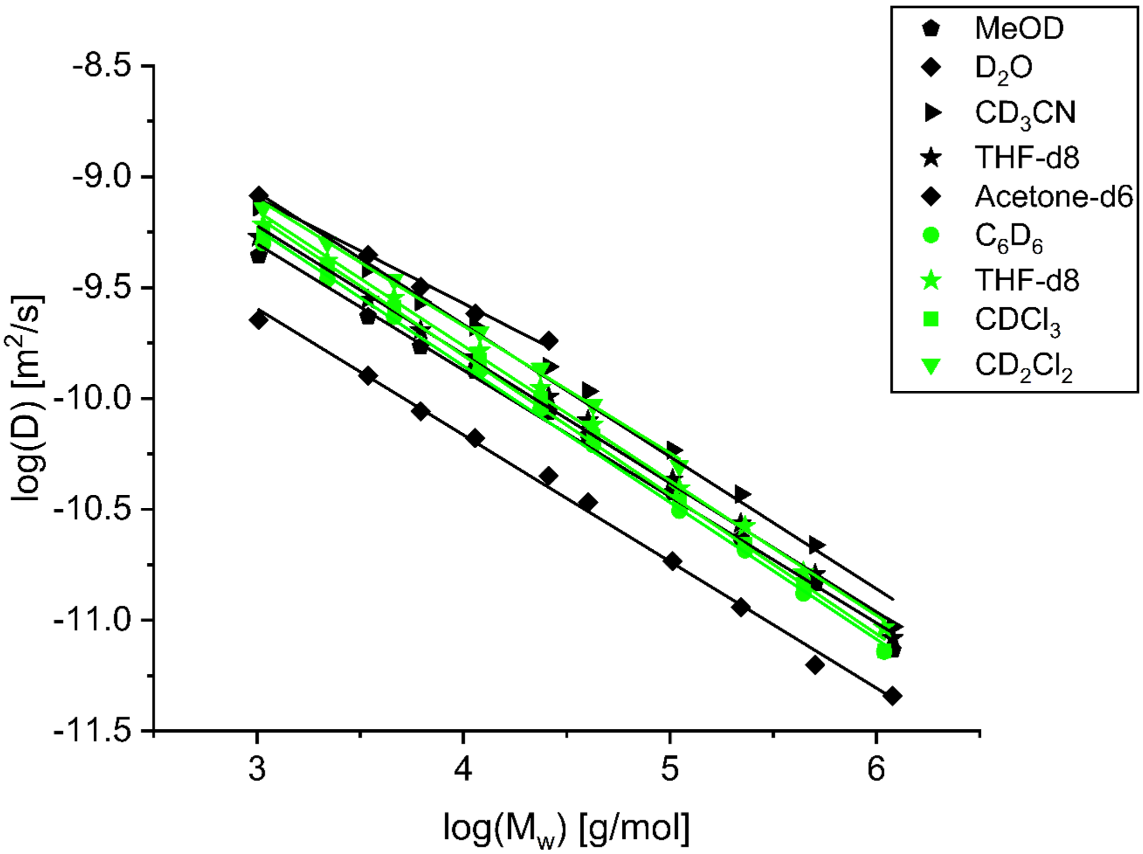

Figure S2. Diffusion coefficients of PEO (black symbols) and PI (green symbols) in dependence of the molar mass for different solvents.

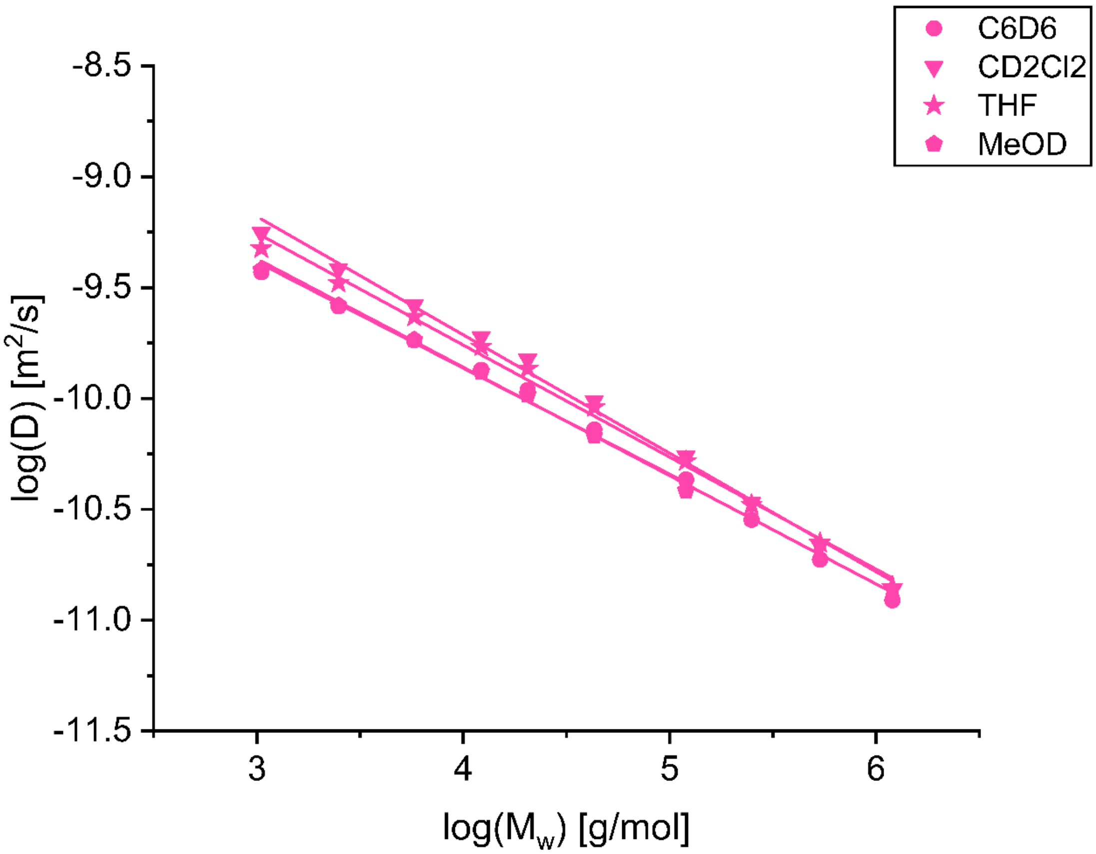

Figure S3. Diffusion coefficients of P2VP in dependence of the molar mass for different solvents.

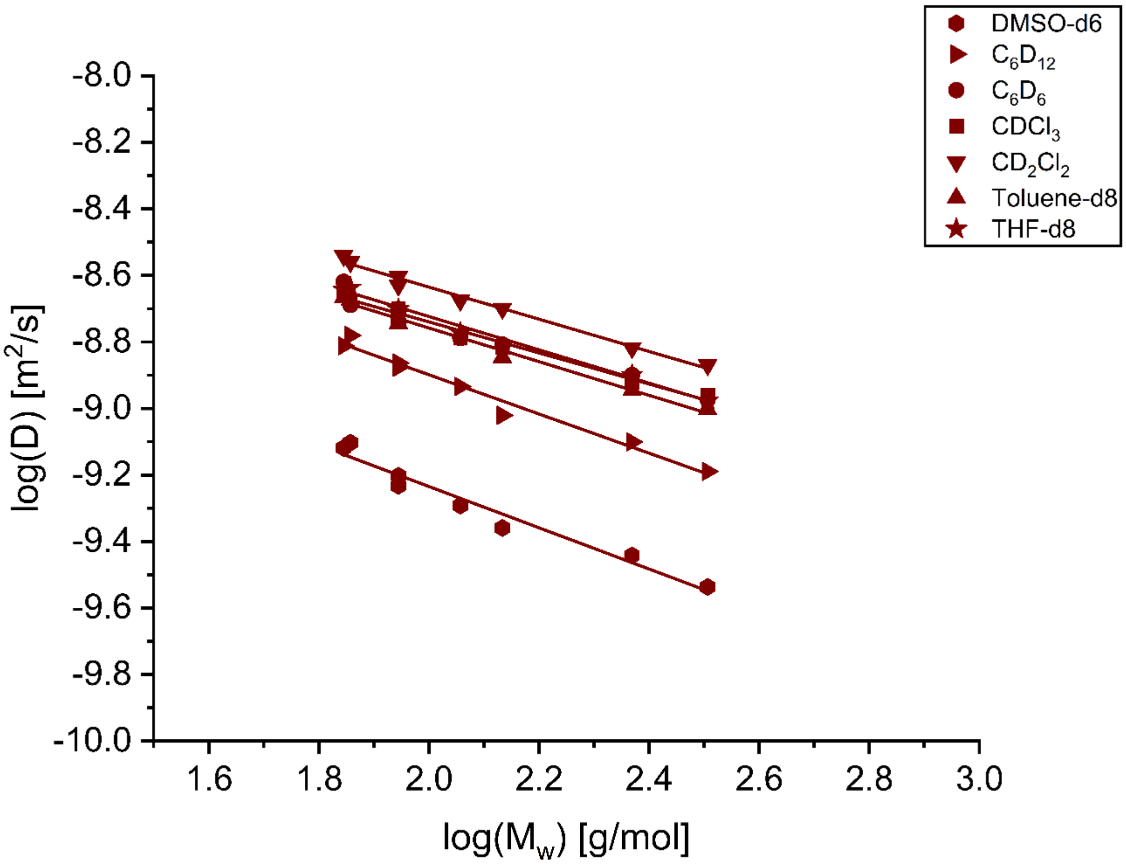

Figure S4: Diffusion coefficients of small molecules (compact spheres) in dependence of the molar mass for different solvents (reproduced from ref (1) with permission of the Royal Society of Chemistry and reproduced from ref (2) with permission of John Wiley and Sons).

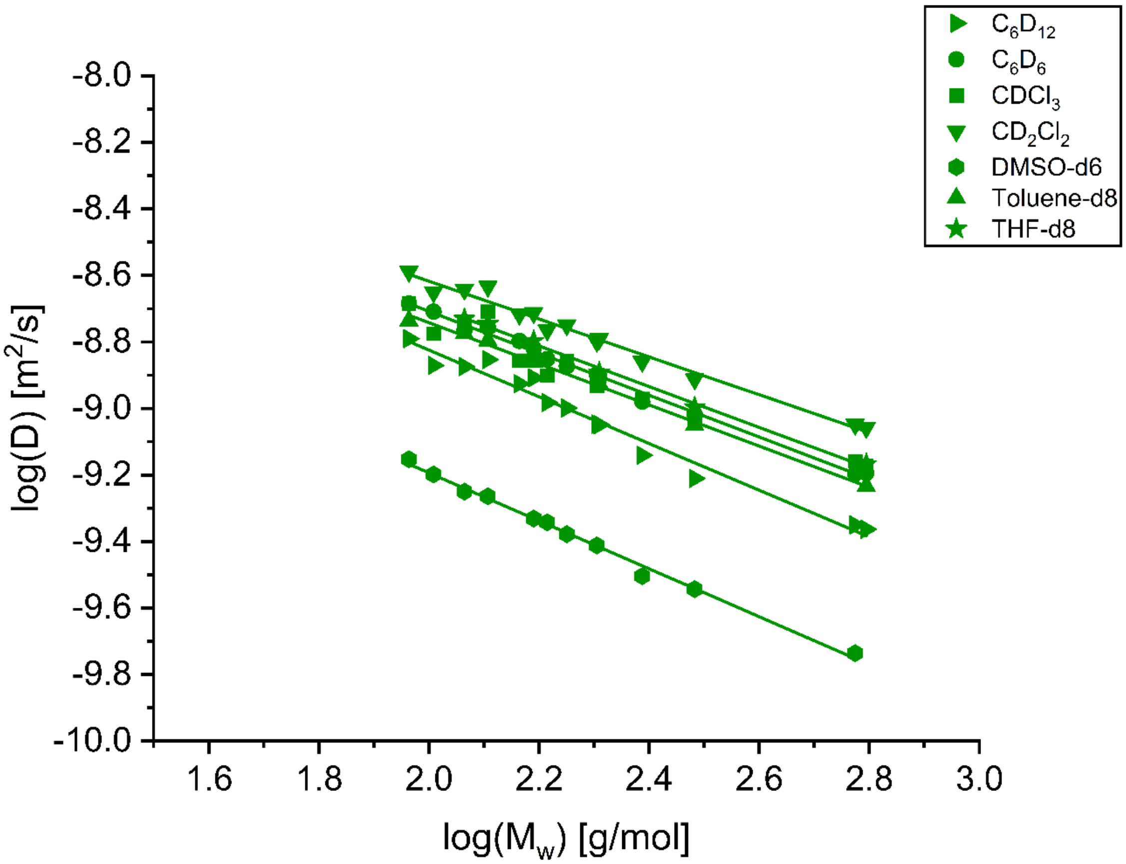

Figure S5: Diffusion coefficients of small molecules (dissipated spheres and ellipsoids) in dependence of the molar mass for different solvents (reproduced from ref (1) with permission of the Royal Society of Chemistry and reproduced from ref (2) with permission of John Wiley and Sons).

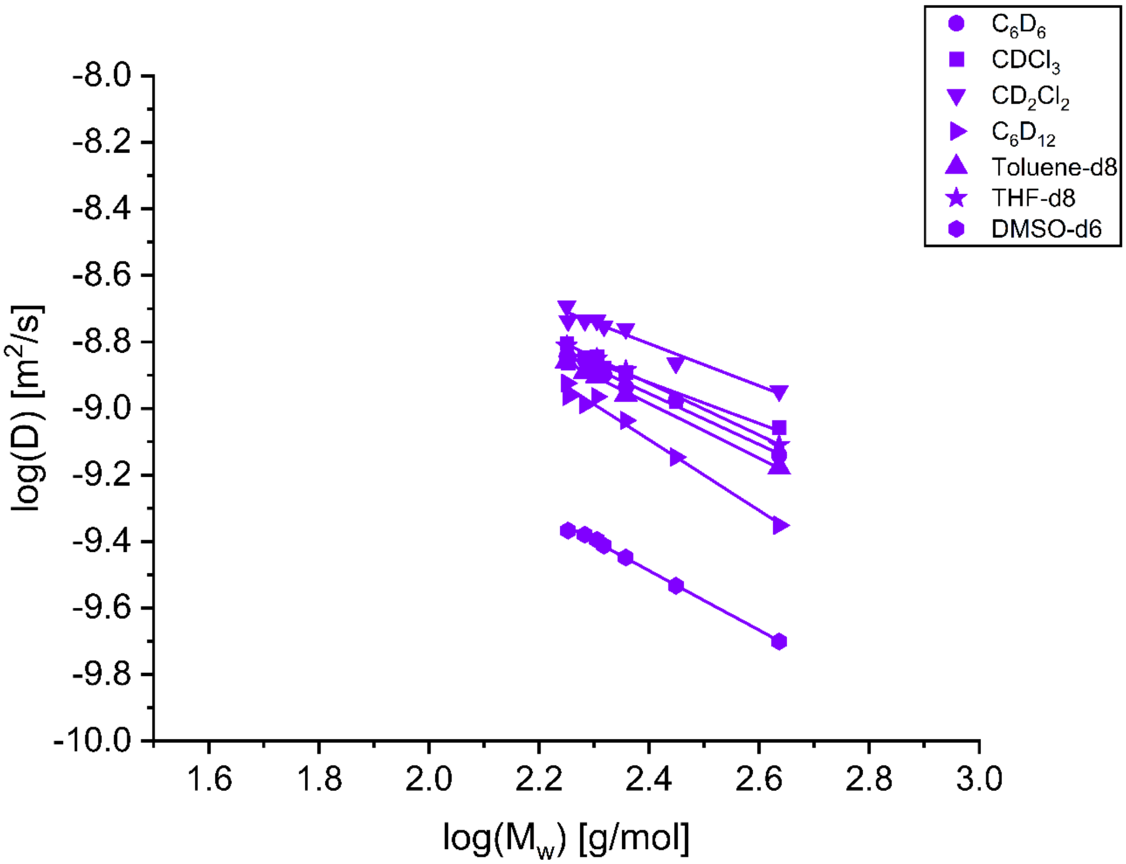

Figure S6. Diffusion coefficients of small molecules (expanded discs) in dependence of the molar mass for different solvents (reproduced from ref (1) with permission of the Royal Society of Chemistry and reproduced from ref (2) with permission of John Wiley and Sons).

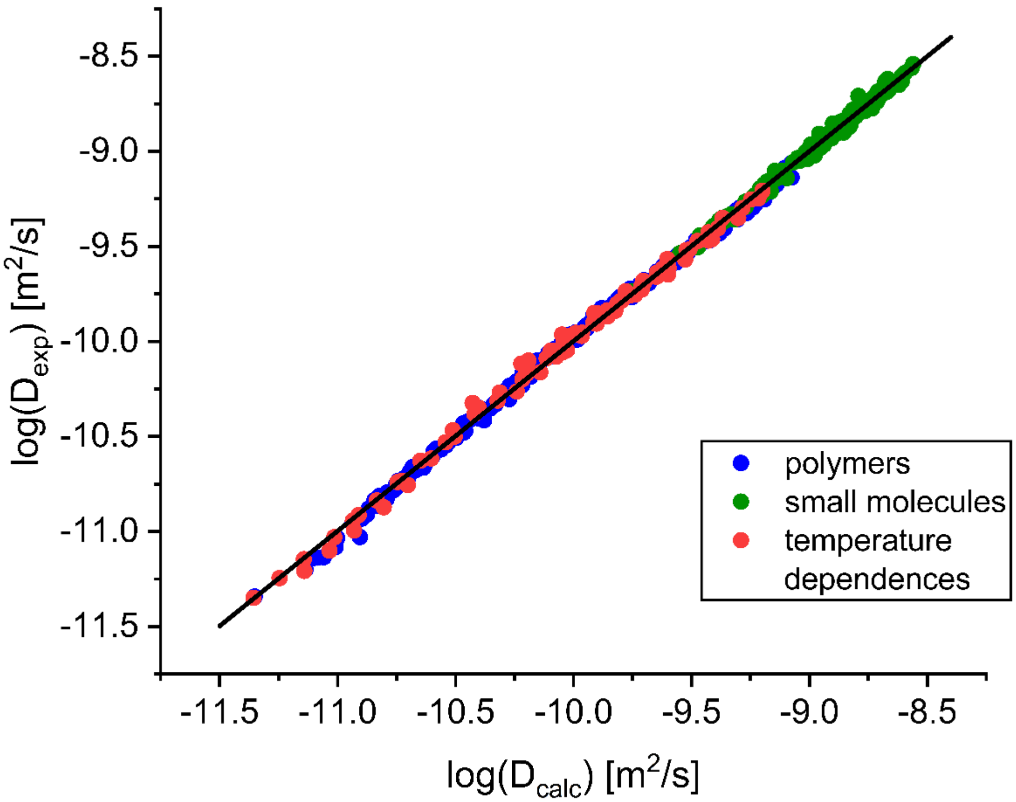

Figure S7: Experimental diffusion coefficients via equation (10) (see main text) vs. calculated diffusion coefficients for all small molecules (green circles), polymers (blue circles) and temperature dependences (red circles), respectively.